Research Evolution — Tesla, Inc. (TSLA), FIN 43900 Project 1

Student: Miles Nine · **Date:** October 19, 2026

> Educational project for FIN 43900. Not investment advice.

Edition A

Edition A was submitted in Brightspace as a link to my GitHub repository, but it was not AI-free, and I am disclosing that here. My Edition A is the research report first committed on **September 3, 2026 at 14:28** (commit `e8e2550`, file `FIN 439/Tesla_2026-09-03_report.md`; an identical copy is at `01_Research/Tesla_2026-09-03_report.md`, verified unchanged with `git diff`). I used **ChatGPT/Codex** to help draft it because I did not know how to write it without AI; the report itself discloses this in its last line. It therefore does not meet the AI-closed rule for Edition A.

Following the course's AI Boundary Card ("If AI was used accidentally or too early"), I have **not changed or recreated it**: it is reproduced unchanged below as my Edition A of record, and I notified a course TA by message on October 19, 2026. The comparison after it uses Edition A as my earliest dated position, with that limitation stated.

Edition A, reproduced unchanged (commit e8e2550, 2026-09-03; heading levels nested for layout only, text unchanged)

Tesla, Inc. (TSLA) — Company Research Report **Valuation date:** September 1, 2026. **Source: UNSOURCED — assignment input.** **Scope:** Filing-based preliminary research; no price target. **Source: UNSOURCED — author scope.** **Primary document:** Tesla, Inc., Form 10-K for the year ended December 31, 2025, filed January 29, 2026, accession `0001628280-26-003952`. **Source: Tesla 2025 Form 10-K, Cover page, p. 1.** #### Executive summary Tesla is a Texas corporation whose common stock trades on the Nasdaq Global Select Market under `TSLA`. **Source: Tesla 2025 Form 10-K, Cover page, p. 1.** The company reports two operating and reportable segments: automotive and energy generation and storage. **Source: Tesla 2025 Form 10-K, Item 1, p. 2.** The investment case has conflicting signals. Automotive revenue declined in 2025, while energy-generation-and-storage revenue grew and Tesla reported substantial cash and short-term investments. **Source: Tesla 2025 Form 10-K, Item 8, Consolidated Statements of Operations, p. 50; Item 8, Consolidated Balance Sheets, p. 49.** Tesla also expects 2026 capital expenditures to exceed $20 billion. **Source: Tesla 2025 Form 10-K, Item 7, MD&A—2026 Outlook, p. 32.** These cross-currents support further research rather than an immediate position decision. **Source: UNSOURCED — preserved author judgment.** #### Business overview Tesla's automotive segment includes the design, development, manufacturing, sales, and leasing of electric vehicles, as well as automotive regulatory credits; the energy-generation-and-storage segment includes the design, manufacture, installation, sales, and leasing of energy products and related services. **Source: Tesla 2025 Form 10-K, Item 1, pp. 2–3; Note 16, Segment Reporting, p. 97.** Powerwall is designed for homes and small commercial facilities, while Megapack serves commercial, industrial, utility, and energy-generation customers. **Source:

Tesla 2025 Form 10-K, Item 1, p. 3.** Tesla also describes investments in AI-enabled products and services, including Robotaxi and Optimus. **Source: Tesla 2025 Form 10-K, Item 7, MD&A—2026 Outlook, pp. 31–32.** The timing, economics, and probability of commercial success for those products remain unresolved in this analysis. **Source: UNSOURCED — preserved author uncertainty.** #### 2025 operating performance Total revenue was $94.827 billion in 2025, down from $97.690 billion in 2024. Automotive revenue was $69.526 billion, down 10% from $77.070 billion, and energy-generation-and-storage revenue was $12.771 billion, up 27% from $10.086 billion. **Source: Tesla 2025 Form 10-K, Item 8, Consolidated Statements of Operations, p. 50; Item 7, MD&A—Results of Operations, pp. 34–37.** Tesla reported 46.7 GWh of energy-storage deployments in 2025. **Source: Tesla 2025 Form 10-K, Item 7, MD&A—2025 Highlights, p. 31.** The company attributed the energy segment's 2025 revenue growth primarily to higher Megapack and Powerwall deployments, partly offset by a lower Megapack average selling price. **Source: Tesla 2025 Form 10-K, Item 7, MD&A—Energy Generation and Storage Segment, p. 37.** Energy-storage deployments and related revenue can be evaluated as a distinct enterprise-value driver rather than presumed to be offset by weaker automotive economics. **Source: UNSOURCED — preserved author judgment.** Automotive regulatory-credit revenue was $1.993 billion in 2025. **Source: Tesla 2025 Form 10-K, Item 8, Consolidated Statements of Operations, p. 50.** Regulatory credits should be separated from recurring automotive operating performance when developing normalized margins. **Source: UNSOURCED — preserved author judgment.** #### Liquidity, debt, and claim hierarchy At December 31, 2025, Tesla reported $16.513 billion of cash and cash equivalents and $27.546 billion of short-term investments. **Source: Tesla 2025 Form 10-K, Item 8, Consolidated Balance Sheets, p. 49; Note 4, Cash and Cash Equivalents, p. 70.** The filing reported $8.177 billion of unpaid debt principal, consisting of $3 million of recourse debt and $8.174 billion of non-recourse debt. **Source: Tesla 2025 Form 10-K, Note 9—Debt, pp. 73–74.** Debt and finance leases had a $8.376 billion net carrying value, split between $1.640 billion current and $6.736 billion long term. **Source: Tesla 2025 Form 10-K, Item 8, Consolidated Balance Sheets, p. 49; Note 9—Debt, p. 73.** The balance sheet also reported $58 million of redeemable noncontrolling interests and $670 million of noncontrolling interests. **Source: Tesla 2025 Form 10-K, Item 8, Consolidated Balance Sheets, p. 49.** A future enterprise-to-equity bridge should explicitly distinguish cash, short-term investments, debt, finance leases, noncontrolling interests, redeemable noncontrolling interests, and other debt-like or non-common claims. **Source: UNSOURCED — preserved author judgment.** Whether short-term investments are fully non-operating, and the appropriate treatment of leases and noncontrolling interests, remain open questions. **Source: UNSOURCED — preserved author uncertainty.** #### Reinvestment and valuation readiness Tesla expects 2026 capital expenditures to exceed $20 billion, driven by AI initiatives, compute infrastructure and data centers, manufacturing and R&D lines and facilities, AI-enabled fleet assets, and its retail, service, and charging footprint. **Source: Tesla 2025 Form 10-K, Item 7, MD&A—Cash Flow and Capital Expenditure Trends, p. 32.** That expected level of capex is material to reinvestment and free-cash-flow forecasts. **Source: UNSOURCED — preserved author judgment.** The split between maintenance and growth capex, and its timing, is not established here. **Source: UNSOURCED — preserved author uncertainty.** The FY2025 diluted weighted-average share count was 3.528 billion, comprising 3.225 billion basic weighted-average shares and 303 million stock-based awards. **Source: Tesla 2025 Form 10-K, Note 4—Earnings per Share, p. 61.** That figure is an initial convention only and must be validated against a valuation-date share-count approach before it is used to calculate per-share value. **Source: UNSOURCED — preserved author judgment.** This report does not treat the class synthetic enterprise-to-equity exercise as a Tesla valuation. **Source: UNSOURCED — preserved author judgment.** Any company-specific valuation should use FCFF and enterprise value before translating to common equity value, with a consistent bridge policy. **Source: UNSOURCED — preserved author judgment.** #### Work required before an initiation decision The next stage should reconcile the balance sheet, debt disclosures, and noncontrolling-interest disclosures into a single claim hierarchy; develop a five-year FCFF forecast; separate recurring results from

regulatory credits and nonrecurring or classification effects; and assess DCF, reverse-DCF, and comparable-company ranges. **Source: UNSOURCED — preserved author research plan.** The decision should be tested under downside cases for automotive revenue, energy-storage growth, operating margin, capital expenditures, and cost of capital. **Source: UNSOURCED — preserved author research plan.** The September 1, 2026 valuation date is after the December 31, 2025 financial-statement date. **Source: UNSOURCED — valuation-date input and date comparison.** Current market price, current capital structure, and a valuation-date share count are therefore not established by the cited 2025 Form 10-K alone. **Source: UNSOURCED — preserved author uncertainty.** #### Investment Decision **Watch / defer.** Do not initiate a Tesla position on this filing-only analysis; the evidence supports deeper valuation work, not a current position decision. **Source: UNSOURCED — preserved author judgment.** This call would change if updated valuation-date evidence and a completed, downside-tested FCFF and enterprise-to-equity bridge supported a position decision. **Source: UNSOURCED — preserved author judgment.** --- This report was written for FIN 43900 (AI Finance Applications, Purdue) as a learning exercise. It is not investment research and it is not financial advice. AI assistance: drafted with ChatGPT/Codex, resumed from my Lab 03 session; sources gathered and verified by me; the judgments are mine. Any remaining errors are my own.

Edition B — current position (October 19, 2026)

Recommendation: WATCH / DEFER. Do not initiate a TSLA position at the current price. The five-year pro-forma model values the equity at about **$27–$29 per share** in the base case (FCFF $27.22, FCFE $29.05), with a scenario range of **$3 (recession) to $88 (bull)**, against a market price of **$378.25** (September 24, 2026). The market price requires either a 3.58% cost of equity, 9.8% perpetual growth, or about $451B of robotaxi revenue by 2030. Full reasoning is in `docs/decision-memo.md`.

What changed from Edition A to Edition B

Changed item	Edition A (Sept 3 report, AI-assisted, disclosed)	Edition B (Oct 19)	Primary driver of change	Evidence or test that justified the change
Thesis / recommendation	"**Watch / defer.** Do not initiate a Tesla position on this filing-only analysis; the evidence supports deeper valuation work, not a current position decision." No value estimate.	**No material change in the action: still WATCH / DEFER**, but now backed by a valuation: base $27–$29, range $3–$88, versus a $378 price.	Own analysis	Completed five-year pro-forma with FCFF and FCFE DCFs; every integrity check passes; reverse DCF shows the price needs $451B of robotaxi revenue by 2030 (tsla-proforma-model/output/visible_o
Data / source choice	FY2025 10-K only; "current market price, current capital structure, and a valuation-date share count are therefore not established."	Three 10-Ks (FY2023–FY2025), the Q2 2026 10-Q, 8-K delivery releases, and a market price with an as-of date ($378.25, Sept 24, 2026). Diluted shares 3,540.0M from the 10-Q.	Course instruction (three filings, as-of dates) and AI (automated XBRL check)	176 stored historical values matched SEC XBRL data with zero mismatches. Model checks pass in every forecast year.

Changed item	Edition A (Sept 3 report, AI-assisted, disclosed)	Edition B (Oct 19)	Primary driver of change	Evidence or test that justified the change
Assumption / model design	Planned "a five-year FCFF forecast." The Lab 06 version used one starting FCFF ($6,433M) with a placeholder 10% WACC, giving **$39.74/share** (Lab 08).	A driver-based, three-statement pro-forma (deliveries × price, energy GWh, services, robotaxi line, capex, working capital), valued with FCFF at a **10.33% WACC** from CAPM and a bottom-up peer beta. Base value **$27.22** (FCFF).	Mixed: course instruction (pro-forma method), AI (code build), own analysis (assumptions and labels)	The capex guidance rose from "**in excess of $20 billion**" (FY2025 10-K, MD&A, p. 32) to "**in excess of $25 billion**" (10-Q Q2 2026, p. 35). This makes FCFE negative in 2026–27, which is the main reason the value fell from $39.74 to about $27–$29.
Risk, limitation, or uncertainty	"The timing, economics, and probability of commercial success for [Robotaxi and Optimus] remain unresolved."	Now **quantified**: robotaxi is an explicit judgment line ($8B by 2030 in base). The market price needs about **$451B**. The terminal value is 110% of DCF value, because 2026–27 cash flow is negative.	Own analysis	Reverse-DCF table and the one-at-a-time tornado (`visible_output.md`); Locked Changed-Input Record: 10% fewer cars cut value by 10.6% with no change in the decision (`docs/validation-and-ai-use.md`).

AI output I checked

Codex's FCFF build (October 19, 2026) — corrected. Codex built the new FCFF/WACC module, but when the tests were run, 11 of 71 failed and the export crashed with `KeyError: 'interest_rate_debt'`. Codex had read the cost of debt as a single setting when the model stores it as a yearly assumption. Codex then fixed it, but reported that it could not run the tests itself. I did not accept that claim on trust: the tests were re-run independently, all **71 passed**, and the export reproduced the FCFF value of **$27.22**. Disposition: **corrected**, then accepted after the independent test run.

What changed from Edition A to Edition B, and why?

My recommendation did not change: I recommended watch/defer on September 3, and that is still my recommendation. What changed is the reason that I found. In September, I deferred because I didn't have enough evidence to value Tesla. Now I defer because I have a tested valuation and model that shows the market price depending on the robotaxi and AI results that Tesla doesn't yet report. The biggest single change in the numbers came from Tesla raising its 2026 capex guidance from over $20 billion to over $25 billion, which turns two years of cash flow negative and lowers the value.